Harjeet Singh Chahal

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Education

Rutgers University - New Brunswick, New Jersey, USA Aug 2025 - May 2027

M.S. in Computer Science **GPA: 4.00/4.00**

- **Relevant Coursework**: Machine Learning, Massive Data Mining, Interpretable & Explainable AI, Database Systems for Data Science, Data Structures & Algorithms, Artificial Intelligence, Visual Computing.

Netaji Subhas University of Technology - New Delhi, India Nov 2020 - Jun 2024

B.Tech. in Computer Science & Engineering (Artificial Intelligence) **GPA: 3.74/4.00**

- **Relevant Coursework**: Deep Learning, Natural Language Processing, High Performance Computing, Distributed Computing, Operating Systems, Data Communication, Game Theory, Design and Analysis of Algorithms, Web Technology.

Technical Skills

Languages: Java, Python, C++, Go, SQL, JavaScript, R, HTML/CSS
ML / AI: PyTorch, Scikit-Learn, Hugging Face Transformers, NLTK, LangChain, RAG, OpenCV, NumPy, Pandas
Backend / Frontend: FastAPI, Flask, Node.js, REST, GraphQL, gRPC, MCP, React, React Native, Next.js, Expo
Cloud & DevOps: AWS, GCP, Docker, Kubernetes, Linux, Git, CI/CD, SLURM, Prometheus, Grafana
Data Tools: PostgreSQL, pgvector, MySQL, MongoDB, Redis, FAISS, Pinecone, ChromaDB, Kafka

Experience

Rutgers School of Health Professions Piscataway, NJ

Research Assistant : Mobile + ML Engineering Apr 2026 - Present

- Productionized HIPAA-compliant PyTorch EfficientNet-B0 on FastAPI, cutting CPU inference to **25–40 ms**.
- Deployed Dockerized FastAPI ML APIs on **GCP Cloud Run** with **CI/CD** for automated tests, builds, and releases.
- Built a 50-test Python/FastAPI suite for ML inference, auth, edge cases, and parity; resolved ARM/x86 floating-point drift and regressions and owned **10+ Apple TestFlight releases** tested by **100+ participants**.

Technical Consulting and Research Inc. Weston, CT

Software Development Intern : Backend + AI Engineering Jun 2026 - Aug 2026

- Architected a **local-first** compliance platform (FastAPI, React, PostgreSQL/pgvector) running LLM inference and embeddings on-premise via **Ollama**, keeping client evidence off third-party APIs.
- Built a pipeline restricting citations to retrieved chunk IDs, with server-side source resolution to reduce unsupported claims.
- Developed a **MCP** server exposing 20 read-only compliance tools for agentic retrieval, evidence lookup and control analysis.

Dhruv Research Gurugram, India

Associate : Analytics + Data Engineering Sep 2024 – May 2025

- Engineered Python/SQL ETL pipelines with Pandas/NumPy to clean, validate, and normalize **200K+ survey responses** across 10 state projects, automating QA and transformations to **reduce turnaround time by 30%**.
- Predicted vote share across **500+ constituencies** using Python/SQL, statistical weighting, and MAE/RMSE validation.
- Worked within and coordinated an **8-member team**, using Python/SQL trackers for sampling and demographic skew.

Projects

ScaleTrain : HPC Training Pipeline | PyTorch, DDP, SLURM, NVIDIA A100 | GitHub May 2026 - July 2026

- Built fault-tolerant PyTorch DDP training on SLURM, scaling a **30M-param** transformer to **4 A100s at 500K tokens/s**.
- Engineered signal-driven checkpoints and deterministic data partitioning for **exact resume with zero lost work**.
- Automated SLURM setup, fault injection, benchmarking, metrics, and reporting in an idempotent pipeline with **64 tests**.

GraphRec | Python, SciPy, Streamlit, Docker | *GitHub* Sept 2025 - Dec 2025

- Built a Personalized **PageRank** recommender on a 31.8M-rating bipartite graph of 200,948 users using sparse matrices.
- Fixed a dimension mismatch bug crashing PageRank and **cut 25 GB** of per-query dense memory allocation.
- Achieved **0.280 Hit Rate@10** vs **0.160** for collaborative filtering baseline, backed by 197 pytest tests.

ConvLoRA: Parameter-Efficient CNN Transfer Learning | PyTorch, ResNet-18 | *GitHub* Jan 2026 - May 2026

- Implemented **convolutional LoRA from scratch** in PyTorch, adapting ResNet-18 for CIFAR-10 to CIFAR-100 transfer.
- Achieved **59.34% CIFAR-100 accuracy** using 337K trainable parameters, just **3.01%** of the parameters in full fine-tuning.
- Benchmarked LoRA ranks 1-32 versus fine-tuning & linear probing, cutting CIFAR-10 forgetting from **5.06 to 1.98 points**.

Customer Churn Prediction | MLOps, DVC, MLflow, Docker | GitHub Apr 2024 – June 2024

- Built an MLOps pipeline for churn prediction (7K records): MLflow tracking, **containerized FastAPI on Render**, CI/CD.
- Benchmarked **logistic regression vs. XGBoost and Random Forest**; shipped the simplest at **84.0%** ROC-AUC.
- Fixed a silent train/serve skew that made the production API ignore a feature; fixed it and added **30 regression tests**.

Achievements

Programming: Solved 1,550+ LeetCode problems with a 1,300+ day active streak. | leetcode.com/u/harjeetchahal
Academic Distinction: Ranked top 0.15% nationally among ∼1M candidates in JEE Main 2020; GRE Quant: 166/170.